

Tema d'esame 26/09/2017

$$\int_0^{\pi} dx \frac{\cos(3x)}{5+4\cos x}$$

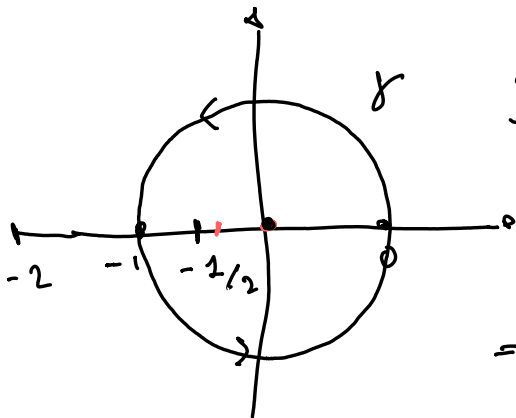
$$\cos x \text{ è pari} \Rightarrow \cos x = \cos(-x)$$

$$\Rightarrow \int_0^{\pi} dx \frac{\cos(3x)}{5+4\cos x} = \int_{-\pi}^0 dx \frac{\cos(3x)}{5+4\cos x} = \frac{1}{2} \int_{-\pi}^{\pi} \frac{\cos 3x}{5+4\cos x} = I$$

Si riporta ad un integrale sulla circonferenza unitaria in $\mathbb{C}$
 cambio di variabile $z = e^{ix}$ $dz = iz dx$

$$2 \cos x = z + \frac{1}{z}$$

$$2 \cos 3x = z^3 + \frac{1}{z^3}$$



$$I = \frac{1}{4i} \int_{-\pi}^{\pi} \frac{z^3 + z^{-3}}{5 + 2(z + z^{-1})} \frac{1}{z} dz$$

$$= \frac{1}{4i} \int_{-\pi}^{\pi} \frac{z^6 + 1}{z^4 (5z + 2z^2 + 1)} dz$$

$$= \frac{1}{4i} \int_{\gamma} \frac{z^6 + 1}{z^3 (2z^2 + 5z + 2)} dz$$

poli : $z=0$ (triplo)

$$\left. \begin{array}{l} z_{\pm} = \frac{-5 \pm \sqrt{25-16}}{4} \\ \quad \quad \quad -1/2 \\ \quad \quad \quad -2 \end{array} \right\} \text{poli semplici}$$

$$I = \left[\text{Res}(f, z=0) + \text{Res}(f, z=-\frac{1}{2}) \right] 2\pi i$$

$$\text{Res}(f, z=-\frac{1}{2}) = \lim_{z \rightarrow -\frac{1}{2}} \frac{1}{4i} \frac{(z+\frac{1}{2}) z^6 + 1}{2 z^3 (z+\frac{1}{2})(z+2)} =$$

$$= \frac{1}{4i} \frac{\left(\frac{1}{2}\right)^6 + 1}{\frac{2}{8} \left(2 - \frac{1}{2}\right)} = \frac{i \frac{2}{3} \left[1 + \frac{1}{64} \right]} = i \frac{2}{3} \left[\frac{65}{64} \right] = i \frac{65}{96}$$

in $z=0$ ha un polo di ordine 3

$\Rightarrow$ sviluppo intorno a $z=0$

$$f(z) \propto \frac{z^6 + 1}{z^3 (2z^2 + 5z + 2)} = \frac{1 + z^6}{2z^3 (1 + \frac{5}{2}z + z^2)}$$

devo sviluppare (...) al secondo ordine per avere il termine $\propto \frac{1}{z}$

$$(1+z)^{-1} \sim 1 - z + z^2 + o(z^2)$$

$$= \frac{1+z^6}{2z^3} \left(1 - \frac{5}{2}z - z^2 + \left(\frac{5}{2}z + z^2\right)^2 + o(z^2) \right)$$

$$= \frac{1+z^6}{2z^3} \left(1 - \frac{5}{2}z + \left(-1 + \frac{25}{4}\right)z^2 + o(z^2) \right)$$

$$= \frac{1}{2z^3} \left(1 - \frac{5}{2}z + \left(-1 + \frac{25}{4}\right)z^2 + o(z^2) \right)$$

$$= \frac{1}{2z^3} + \frac{5}{4z^2} + \frac{21}{8} \cdot \frac{1}{z} + \text{P.R.} \Rightarrow \text{Res}(f, z=0)$$

$$= \frac{21}{8} \cdot \frac{1}{4i} = -i \frac{21}{32}$$

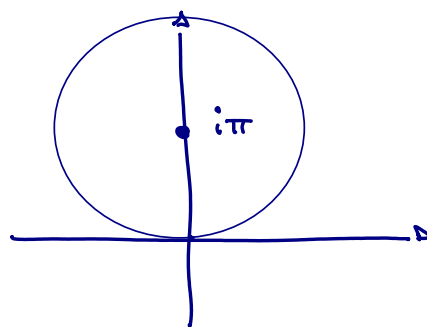
$$\Rightarrow \mathbb{I} = 2\pi i \left[\frac{65}{3 \cdot 32} i - \frac{21}{32} i \right] = \pi \frac{63 - 65}{3 \cdot 32} = -\frac{4\pi}{96}$$

$$= -\frac{\pi}{24} .$$

Tema d'esame 29/01/2019

Sia $\mathcal{C}$ la circonferenza di raggio π e centro $i\pi$.
Calcolare

$$\oint_{\mathcal{C}} dz \frac{e^{z/2} + |z|^2}{(z - i\pi)^2}$$



Non possiamo usare il teorema dei residui perché
la funzione non è olomorfa per via di $|z|^2 = z \cdot \bar{z}$.

Scriviamo

$$\oint_{\mathcal{C}} f(z) dz = \oint_{\mathcal{C}} \frac{dz e^{z/2}}{(z - i\pi)^2} \quad \text{|| } f_1 \quad + \quad \oint_{\mathcal{C}} \frac{dz |z|^2}{(z - i\pi)^2} \quad \text{|| } f_2$$

$\textcircled{\text{I}}$

$\textcircled{\text{II}}$

$\textcircled{\text{I}} \rightarrow f$ analitica $\Rightarrow$ possiamo usare il Teo dei residui.

Abbiamo un polo di ordine 2 in $z = i\pi$

$$\Rightarrow \oint_{\mathcal{C}} f_1(z) dz = 2\pi i \operatorname{Res}(f_1, i\pi)$$

Come calcoliamo $\text{Res}(f_1, i\pi)$?

2 modi

① Sviluppiamo z intorno a $i\pi$

$\Rightarrow w \equiv z - i\pi$ intorno a 0

$$\begin{aligned} \frac{e^{\frac{w+i\pi}{2}}}{w^2} &= e^{i\pi/2} \frac{1}{w^2} e^{\frac{w}{2}} = e^{i\pi/2} \frac{1}{w^2} \sum_{n=0}^{+\infty} \frac{1}{n!} \left(\frac{w}{2}\right)^n \\ &= e^{i\pi/2} \sum_{n=0}^{+\infty} \frac{1}{n!} (w)^{n-2} \cdot \left(\frac{1}{2}\right)^n \end{aligned}$$

mi interessa il coefficiente del termine $\propto \frac{1}{w}$

$$(n=1) \Rightarrow \text{Res}(f_1, i\pi) = e^{i\pi/2} \cdot \frac{1}{2} = \frac{i}{2}.$$

② uso la formula per poli di ordine k in z_0

$$\text{Res}(f, z_0) = \frac{1}{(k-1)!} \left(\frac{\partial}{\partial z} \right)^{k-1} \left((z-z_0)^k f(z) \right) \Big|_{z=z_0}$$

$$\begin{aligned} \text{Res}(f_1, i\pi) &\stackrel{k=2}{=} \frac{1}{1!} \frac{\partial}{\partial z} \left[(z-i\pi)^2 \frac{e^{z/2}}{(z-i\pi)^2} \right] \Big|_{z=i\pi} \\ &= \frac{1}{2} e^{z/2} \Big|_{z=i\pi} = \frac{i}{2}. \end{aligned}$$

$$\Rightarrow \oint_{\gamma} f_1 dz = 2\pi i \cdot \frac{i}{2} = -\pi.$$

Per calcolare la seconda parte facciamo un cambio di variabile

$$\oint_{\mathcal{C}} \frac{z \bar{z}}{(z - i\pi)^2} dz \xrightarrow{\omega = z - i\pi} \oint_{\mathcal{C}'} \frac{(\bar{\omega} - i\pi)(\omega + i\pi)}{\omega^2} d\omega$$

dove $\mathcal{C}'$ è la circonferenza centrata in 0 di raggio π
 $\pi e^{i\theta} \quad \theta \in [0; 2\pi)$

$$\oint_{\mathcal{C}'} \frac{(\bar{\omega} - i\pi)(\omega + i\pi)}{\omega^2} d\omega = \oint_{\mathcal{C}'} \frac{\bar{\omega}\omega - i\pi\omega + i\pi\bar{\omega} + \pi^2}{\omega^2} d\omega$$

$$\omega = \pi e^{+i\theta}, \quad \bar{\omega} = \pi e^{-i\theta} \quad \omega \bar{\omega} = \pi^2$$

$$= \oint_{\mathcal{C}'} \frac{1}{\omega^2} \left[\underbrace{\pi^2}_{\propto \frac{1}{\omega^2} \text{ integrale nullo su } \mathcal{C}'} - i\pi\omega + \underbrace{\frac{i\pi}{\omega}}_{\propto \frac{1}{\omega^3} \text{ integrale nullo su } \mathcal{C}'} + \underbrace{\pi^2}_{\propto \frac{1}{\omega^2} \text{ integrale nullo su } \mathcal{C}'}} \right]$$

$$= \oint_{\mathcal{C}'} -\frac{i\pi}{\omega} = 2\pi i (-i\pi) = 2\pi^2.$$

$$\Rightarrow \boxed{\oint_{\mathcal{C}} f = -\pi + 2\pi^2}.$$

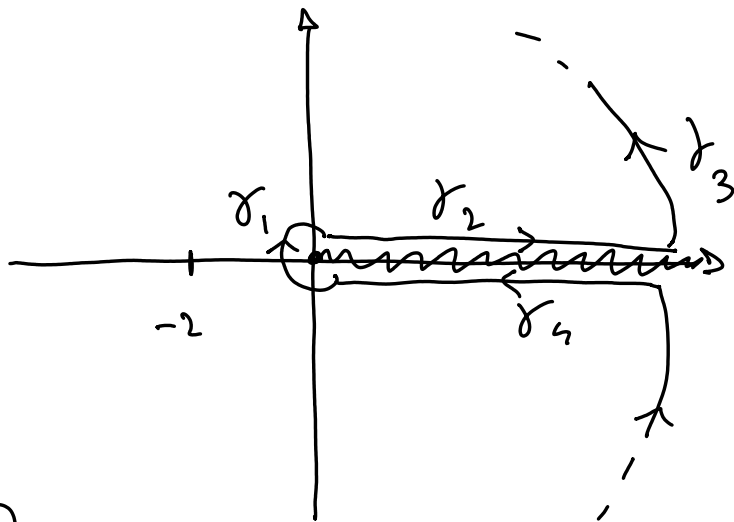
Tema d'esame 19/02/2019

Calcolare $\int_0^{+\infty} dx \frac{1}{x^{2/3}(x+2)}$

Considero $f(z) = \frac{1}{z^{2/3}(z+2)}$

ha un polo semplice in $z = -2$

$z=0$ è una singolarità non isolata (è un Branch point).



① $\gamma_1 = \rho e^{i\theta} \quad \theta \in (-2\pi; 0) \quad \text{con } \rho \rightarrow 0$

$$\lim_{z \rightarrow 0} z \cdot f(z) = \lim_{z \rightarrow 0} z^{1/3} / (z+2) = 0$$

$\Rightarrow$ Per il lemma del cerchio piccolo

$$\oint_{\gamma_1} f(z) dz = 0$$

$$(2) \quad r_2: z = x e^{i \cdot 0} = x$$

$$x \in (p, R) \quad \text{con} \quad p \rightarrow 0 \quad \text{e} \quad R \rightarrow +\infty$$

$$\lim_{\substack{R \rightarrow +\infty \\ p \rightarrow 0}} \int_p^R f(z) dz = \int_0^{+\infty} dx \frac{1}{x^{2/3}(x+2)} = I$$

$$(3) \quad \gamma_3: R e^{i\theta} \quad R \rightarrow +\infty \quad \theta \in (0; 2\pi)$$

$$\lim_{|z| \rightarrow +\infty} z f(z) = \lim_{|z| \rightarrow +\infty} \frac{z^{1/3}}{z+2} = 0$$

$$\hookrightarrow \sim \frac{1}{z^{2/3}} \rightarrow 0$$

$\Rightarrow$ Lemma del cerchio grande

$$\oint_{\gamma_3} f(z) dz = 0$$

$$(4) \quad z = x e^{i \frac{2\pi}{3}}$$

perché
sotto il Branch cut

$$z^{2/3} = x^{2/3} e^{\frac{4}{3}i\pi}$$

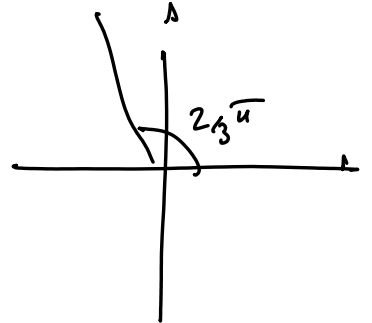
$$\int_{r_4} f(z) dz = \int_{+\infty}^0 dx \frac{1}{e^{\frac{4}{3}i\pi} x^{2/3}(x+2)} = -\frac{1}{e^{\frac{4}{3}i\pi}} I$$

$$\Rightarrow (1 - e^{-\frac{4}{3}i\pi}) I = 2\pi i \operatorname{Res}(f, -2)$$

$$\Rightarrow (1 - e^{-\frac{4}{3}\pi i}) I = 2\pi i \operatorname{Res}(f, -2) =$$

$$= 2\pi i \lim_{z \rightarrow -2} (z+2) \frac{1}{z^{2/3}(z+2)} = \frac{1}{2^{2/3}} \frac{1}{e^{i\pi \cdot 2/3}} \cdot 2\pi i$$

$$\Rightarrow \boxed{I} = \frac{2\pi i}{2^{2/3}} \frac{e^{-i\frac{2}{3}\pi}}{1 - e^{-i\frac{4}{3}\pi}}$$



$$= \frac{2\pi i}{2^{2/3}} \frac{1}{e^{\frac{2}{3}\pi i} - e^{-\frac{2}{3}\pi i}} = \sin \frac{2}{3}\pi = \frac{\sqrt{3}}{2}$$

$$= \frac{\pi}{2^{2/3}} \cdot \frac{2}{\sqrt{3}} = \pi \frac{\sqrt[3]{2}}{\sqrt{3}}$$

Tema d'esame 29/01/2018

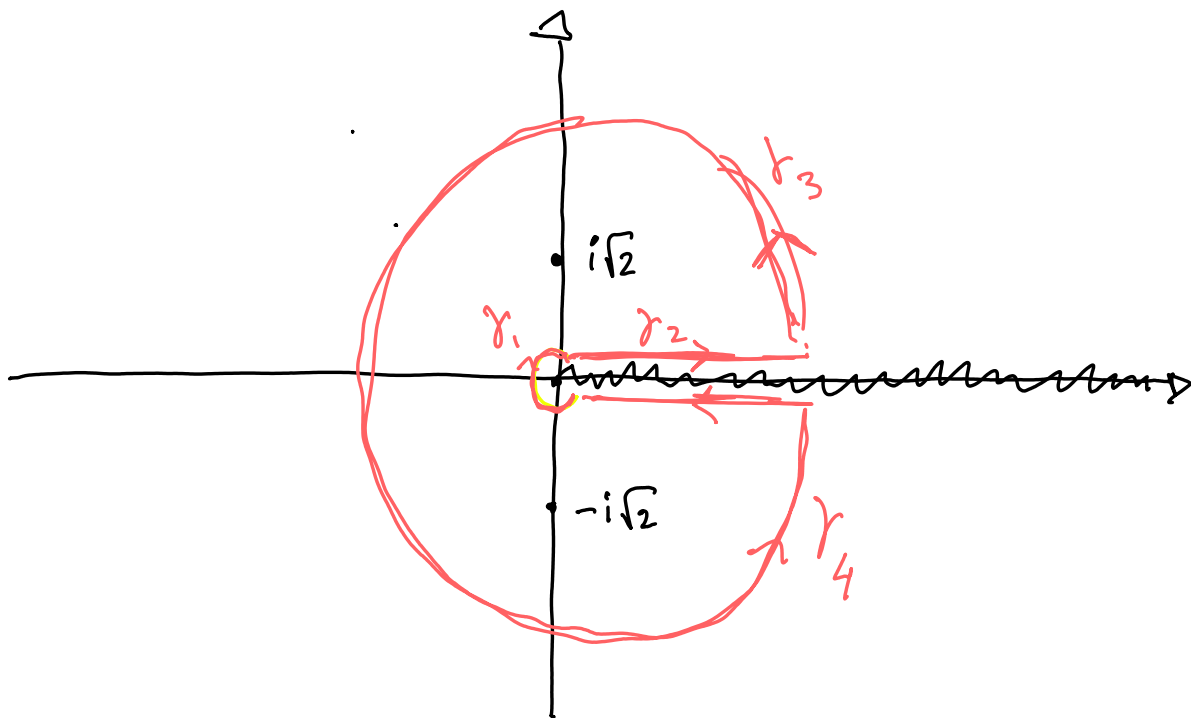
$$\int_0^{\pi/2} \frac{\sqrt{\tan x}}{1 + \cos^2 x} dx$$

poniamo $t = \tan x \rightarrow dt = \frac{dx}{\cos^2 x}$

$$t^2 = \frac{\sin^2 x}{\cos^2 x} = \frac{1 - \cos^2 x}{\cos^2 x} \Rightarrow \cos^2 x = \frac{1}{1 + t^2}$$

$$\Rightarrow \int_0^{+\infty} \frac{\sqrt{t}}{1 + \frac{1}{1+t^2}} \frac{1}{1+t^2} dt = \int_0^{+\infty} \frac{\sqrt{t}}{2+t^2} dt$$

considero $f(z) = \frac{\sqrt{z}}{2+z^2}$ ha poli di ordine 1 in $z = \pm i\sqrt{2}$ e un B.P. in $z = \infty$



$$\begin{aligned}
 \int_{\gamma} f(z) dz &= \int_{\gamma_1 + \gamma_2 + \gamma_3 + \gamma_4} f(z) dz = 2\pi i \operatorname{Res}(f, i\sqrt{2}) + 2\pi i \operatorname{Res}(f, -i\sqrt{2}) \\
 &= 2\pi i \left[\lim_{z \rightarrow i\sqrt{2}} \frac{(z - i\sqrt{2}) \sqrt{z}}{(z - i\sqrt{2})(z + i\sqrt{2})} + \lim_{z \rightarrow -i\sqrt{2}} \frac{(z + i\sqrt{2}) \sqrt{z}}{(z - i\sqrt{2})(z + i\sqrt{2})} \right] \\
 &= \cancel{2\pi i} \left[\frac{\sqrt{2} e^{i\frac{\pi}{4}}}{\cancel{2i\sqrt{2}}} + \frac{\sqrt{2} e^{i\frac{3\pi}{4}}}{\cancel{-2i\sqrt{2}}} \right] \quad \begin{aligned} +i\sqrt{2} &= \sqrt{2} e^{i\frac{\pi}{2}} \\ -i\sqrt{2} &= \sqrt{2} e^{i\frac{3\pi}{2}} \\ e^{i\frac{\pi}{4}} &= \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} \\ e^{i\frac{3\pi}{4}} &= -\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} \end{aligned} \\
 &= \frac{\pi}{\sqrt{2}} \sqrt{2} = \sqrt{2} \cdot \pi
 \end{aligned}$$

$$\textcircled{\text{I}} \int_{\gamma_1} f(z) dz \quad z f(z) = \frac{z \sqrt{z}}{2 + z^2} \sim z \sqrt{z} \rightarrow 0 \text{ per } |z| \rightarrow 0$$

$\Rightarrow$ Per il lemma del cerchio piccolo $\int_{\gamma_1} f(z) dz = 0$

$$\textcircled{\text{II}} \int_{\gamma_2} f(z) dz = \int_0^1 f(x) dx = \text{I}$$

$$\textcircled{\text{III}} \int_{\gamma_3} f(z) dz \quad z f(z) = \frac{z \sqrt{z}}{2 + z^2} \sim \frac{z \sqrt{z}}{z^2} \sim \frac{1}{\sqrt{z}} \rightarrow 0 \text{ per } |z| \rightarrow +\infty$$

$\Rightarrow$ per il lemma del cerchio grande $\int_{\gamma_3} f(z) dz = 0$

(IV)

$$\int_{\gamma_n} f(z) dz$$

sono sotto il branch cut $\Rightarrow \sqrt{z} = -\sqrt{x}$

$$= \int_{+\infty}^0 \frac{-\sqrt{x}}{x^2 + 2} dx = \int_0^{+\infty} \frac{\sqrt{x}}{x^2 + 2} = I$$

$$\Rightarrow 2I = \sqrt[4]{2} \pi \Rightarrow I = 2^{-\frac{3}{4}} \pi \quad \checkmark$$

Tema d'esame 31/01/2023 es. 2

$$f(z) = \frac{1}{z} + \sum_{k=1}^{+\infty} \frac{k}{\pi^k} z^k$$

- determinare dove f è olomorfa
- determinare la funzione calcolando la serie
- calcolare l'integrale sulle curve $C: |z-i|=2$

$$I = \oint_C \frac{dz}{2\pi i} \frac{f(z)}{(z+2)(z-i)}$$

a)

$\frac{1}{z}$ ha un polo semplice in $z=0$

Posso usare il criterio del rapporto per calcolare il raggio di convergenza della serie

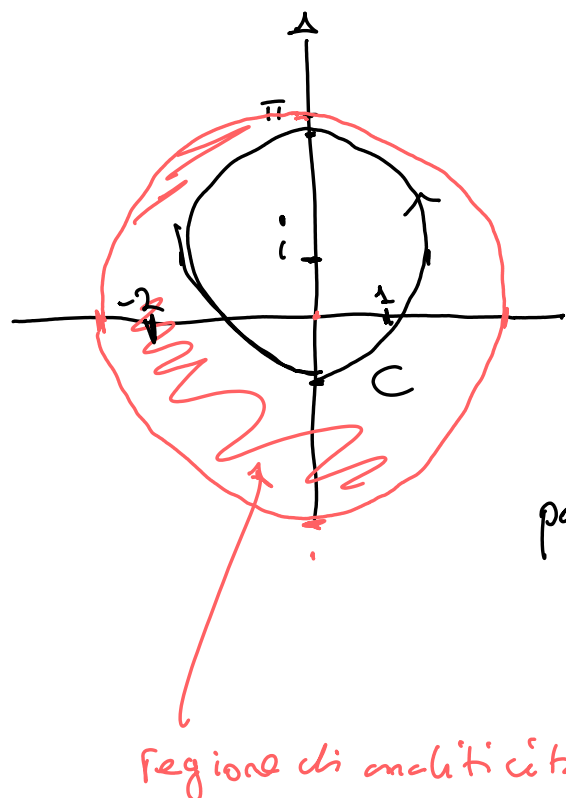
$$R = \lim_{k \rightarrow +\infty} \frac{|a_k|}{|a_{k+1}|} = \lim_{k \rightarrow +\infty} \frac{k}{\cancel{\pi^k}} \frac{\cancel{\pi}^{k+1}}{k+1} = \pi$$

$\Rightarrow f$ olomorfa in $0 < |z| < \pi$

$$\begin{aligned} \text{b) } f(z) &= \frac{1}{z} + z \frac{d}{dz} \sum_{k=0}^{+\infty} \frac{z^k}{\pi^k} = \frac{1}{z} + z \frac{d}{dz} \frac{1}{1 - (z/\pi)} \\ &= \frac{1}{z} + \frac{z\pi}{(\pi - z)^2} \end{aligned}$$

c)

C è contenuta nella regione di analiticità



$$\int_C \frac{dz}{2\pi i} \frac{f(z)}{(z+2)(z-1)}$$

contiene i soli poli $z=0$ e $z=1$ ($|1-i| = \sqrt{2} < 2$)

poli semplici: $z=0 \rightarrow$ dentro C

$z=1$ $|1-i| = \sqrt{2} < 2 \Rightarrow$ dentro C

$z=-2$ $|i+2| = \sqrt{5} > 2 \Rightarrow$ fuori da C

$z=0$ è polo solo del primo addendo, $z=1$ di entrambi

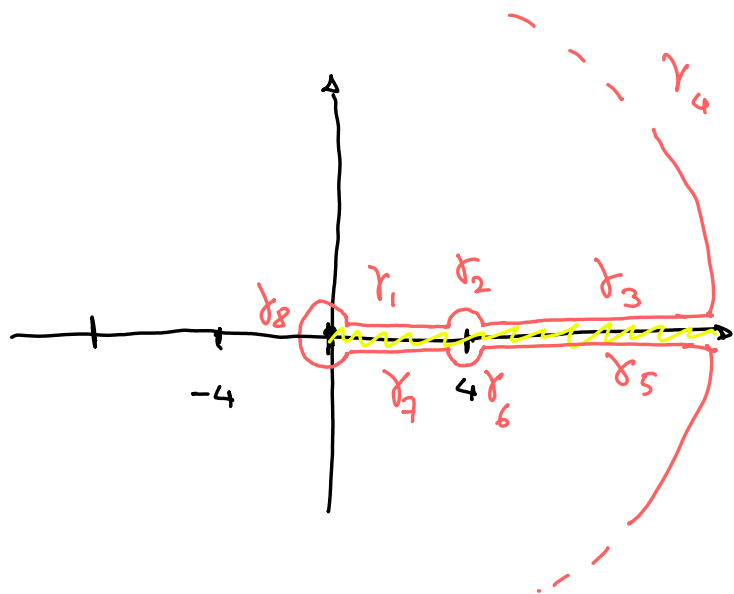
$$I = \text{Res} \left[\frac{1}{z(z+2)(z-1)}, z=0 \right] + \text{Res} \left[\frac{1}{z(z+2)(z-1)}, z=1 \right] \\ + \text{Res} \left[\frac{z\pi}{(z+2)(z-1)(\pi-z)^2}, z=1 \right]$$

$$= \frac{1}{-2} + \frac{1}{3} + \frac{\pi}{3(\pi-1)^2} = -\frac{1}{6} + \frac{\pi}{3(\pi-1)^2}$$

Esercizio

Calcolare
$$\text{PV} \int_0^{+\infty} \frac{\sqrt{x}}{x^2 - 16} dx$$

considero
$$f(z) = \frac{\sqrt{z}}{z^2 - 16}$$



poli semplici in $z = \pm 4$

B.P. in $z = 0$

Nota: l'integrale è da svolgere in parte principale
poiché $f(x)$ non è integrabile in senso classico in
 $z = 4$

$$\gamma_1: z = t + i\varepsilon \quad t \in (r; 4-p) \quad \text{con } \varepsilon, r, p \rightarrow 0$$

$$\gamma_2: z = 4 + pe^{i\theta} + i\varepsilon \quad \theta: \text{da } \pi \text{ a } 0 \quad \varepsilon \rightarrow 0 \quad p \rightarrow 0$$

$$\gamma_3: z = t + i\varepsilon \quad t \in (4+p; R) \quad \varepsilon, p \rightarrow 0 \quad R \rightarrow +\infty$$

$$\gamma_4: z = Re^{i\theta} \quad \theta \in (0; 2\pi) \quad R \rightarrow +\infty$$

$$\gamma_5: z = t - i\varepsilon \quad t \text{ da } R \text{ a } 4+p \quad R \rightarrow +\infty \quad \varepsilon, p \rightarrow 0$$

$$\gamma_6: z = 4 + pe^{i\theta} - i\varepsilon \quad \theta \text{ da } 2\pi \text{ a } \pi \quad p \rightarrow 0 \quad \varepsilon \rightarrow 0$$

$$\gamma_7: z = t - i\varepsilon \quad t \text{ da } 4-p \text{ a } r \quad p, r \rightarrow 0$$

$$\gamma_8: z = re^{i\theta} \quad \theta \text{ da } 2\pi \text{ a } 0 \quad r \rightarrow 0$$

$$1) \lim_{R \rightarrow +\infty} \lim_{\substack{r \rightarrow 0 \\ \rho \rightarrow 0}} \int_{\gamma_1 + \gamma_3} f(z) dz = \text{P.V.} \int_0^{+\infty} f(x) dx$$

$$2) \lim_{r \rightarrow 0} \int_{\gamma_2} f(z) dz = -i\pi \text{Res}(f, 4) = -\frac{i\pi}{4}$$

$\uparrow$
 lemma dei piccoli cerchi $f(z)$ ha polo semplice in $z=4$

$$\text{Res}(f, 4) = \lim_{z \rightarrow 4} (z-4) \frac{\sqrt{z}}{(z+4)(z-4)} = \frac{1}{4}$$

$$3) \lim_{R \rightarrow +\infty} \int_{\gamma_4} f(z) dz = \lim_{R \rightarrow +\infty} \int_0^{2\pi} \frac{\sqrt{R} e^{i\theta/2}}{R^2 e^{i2\theta} - 16} iR e^{i\theta} d\theta$$

$$zf(z) \sim z \frac{\sqrt{z}}{z^2} \sim \frac{1}{\sqrt{z}} \rightarrow 0$$

per $|z| \rightarrow +\infty$

$$\left(zf(z) \sim \frac{z\sqrt{z}}{z^2} \rightarrow 0 \right)$$

lemma del cerchio grande

$$\Rightarrow \text{per il lemma del cerchio grande } \int_{\gamma_4} f(z) dz = 0$$

$$4) \lim_{R \rightarrow +\infty} \lim_{r \rightarrow 0} \lim_{\rho \rightarrow 0} \oint_{\gamma_5 + \gamma_7} f(z) dz = PV \int_{-\infty}^{+\infty} \frac{\sqrt{x}}{x^2 - 16} dx = PV \int_0^{+\infty} \frac{\sqrt{x}}{x^2 - 16} dx$$

perché siamo sotto al taglio
 $\Rightarrow z = Re^{i2\pi}$

$$5) \lim_{r \rightarrow 0} \oint_{\gamma_6} f(z) dz = -i\pi \operatorname{Res}(f, 4) = \frac{i\pi}{4}$$

sempre perché sotto al B.C.

$$6) \lim_{r \rightarrow 0} \oint_{\gamma_8} f(z) dz = \lim_{r \rightarrow 0} \int_0^{2\pi} \frac{\sqrt{r} e^{i\theta/2}}{r^2 e^{2i\theta} - 16} i r e^{i\theta} d\theta = 0$$

$$\left| \frac{\sqrt{r} e^{i\theta/2}}{r^2 e^{2i\theta} - 16} i r e^{i\theta} \right| \leq r^{3/2} \rightarrow 0$$

analogo a
 $zf(z) \sim z\sqrt{z} \rightarrow 0$
 per $|z| \rightarrow 0$
 piccoli cerchi.

Per cui

$$\oint_{\gamma} f(z) dz = 2 \cdot PV \int_0^{+\infty} \frac{\sqrt{x}}{x^2 - 16} dx + \frac{i\pi}{4} - \frac{i\pi}{4}$$

$$= 2\pi i \operatorname{Res}(f, -4) = 2\pi i \lim_{z \rightarrow -4} \frac{\cancel{(z+4)} \sqrt{z}}{\cancel{(z+4)}(z-4)} = 2\pi i \left(-\frac{2i}{8} \right)$$

$$= 2\pi i \left(-\frac{i}{4} \right) = \frac{\pi}{2}$$

$$\Rightarrow PV \int_0^{+\infty} \frac{\sqrt{x}}{x^2 - 16} dx = \frac{\pi}{4}$$

Tema d'esame 9/07/2024

Si trovi il valore principale dei seguenti integrali dove $a \in \mathbb{R}$, $m > 0$

$$C_m(a) = \int_{-\infty}^{+\infty} dx \frac{\cos(mx)}{x-a}, \quad S_m(a) = \int_{-\infty}^{+\infty} dx \frac{\sin(mx)}{x-a}$$

Gli integrali sono parte reale e immaginaria di

$$F_m(a) = C_m(a) + i S_m(a) = \int_{-\infty}^{+\infty} \frac{e^{imx}}{x-a} dx$$

$$\begin{aligned} t &= m(x-a) \\ dt &= m dx \end{aligned} \quad = \int_{-\infty}^{+\infty} \frac{e^{it + ima}}{t/m} \frac{dt}{m}$$

$$= e^{ima} \int_{-\infty}^{+\infty} \frac{e^{it}}{t} dt$$

$$= e^{ima} \int_{-\infty}^{+\infty} \left(\frac{\cos x}{x} + i \frac{\sin x}{x} \right) dx$$

perché dispari

Usando l'integrale notevole

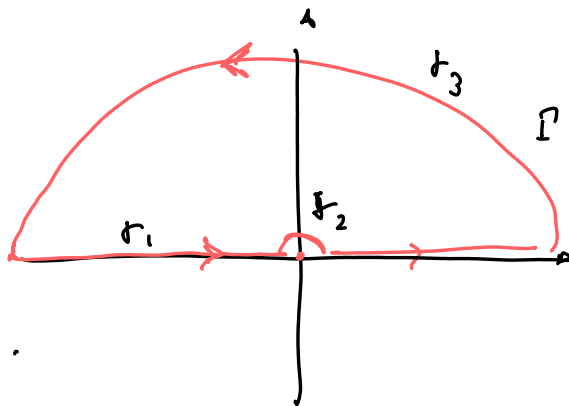
$$\int_{-\infty}^{+\infty} \frac{\sin x}{x} dx = \pi$$

$$= i e^{ima} \int_{-\infty}^{+\infty} \frac{\sin x}{x} dx = i e^{ima} \pi = -\operatorname{Sh}(ma)\pi + i \cos(ma)\pi$$

$$C_m(a) = -\pi \operatorname{Sh}(ma) \quad S_m(a) = \cos(ma)\pi$$

DIMOSTRIAMO L'INTEGRALE NOTEVOLE

consideriamo $\int_{-\infty}^{+\infty} \frac{e^{iz}}{z} dz$



$\frac{e^{iz}}{z}$ ha un polo semplice in 0.

devo chiudere nell'UP $\Rightarrow e^{\lambda z}$ in integrale tipo $e^{i\lambda z} f(z)$ con $\lambda > 0$

Per r_3 uso il lemma di Jordan $\Rightarrow \int_{r_3} f = 0$

Per r_2 : $\int_{r_2} f(z) dz = (0 - \pi) i \operatorname{Res}(f, 0)$
 usando il lemma del cerchio piccolo
 $= -\pi i \lim_{z \rightarrow 0} e^{iz} = -\pi i$

$\int_{r_1} f(x) dx = P \int_{-\infty}^{+\infty} \frac{e^{ix}}{x} dx$

$\Rightarrow 0 = P \int_{-\infty}^{+\infty} \frac{e^{ix}}{x} dx - \pi i \Rightarrow P \int_{-\infty}^{+\infty} \frac{e^{ix}}{x} dx = \pi i$

separando parte reale
e parte immaginaria

$\Rightarrow \int_{-\infty}^{+\infty} \frac{\sin x}{x} dx = \pi$